AI DOCUMENT ASSISTANT

Demonstration Test Report

A structured sample for testing document ingestion, retrieval-augmented generation, and source-grounded answers.

Document property	Value
Project name	Orion Knowledge Hub
Document version	1.0
Prepared for	AI Document Assistant demonstration
Document date	23 October 2026
Document type	Fictional project report for software testing

Purpose of this test document

This report includes factual statements, headings, lists, and tables. Use it to verify text extraction, indexing, retrieval, grounded answers, source attribution, follow-up questions, and safe handling of information absent from the source.

All people, budgets, and project details in this report are fictional.

1. Executive Summary

Orion Knowledge Hub: project overview

Orion Knowledge Hub is a fictional internal knowledge platform designed to help a small team organize project documentation and locate verified information. The pilot uses a document-grounded AI assistant so team members can ask natural-language questions about approved files.

The pilot is planned for six weeks, beginning on 2 November 2026 and ending on 11 December 2026. The project budget is INR 450,000. The project owner is Mira Shah, and the technical lead is Arjun Mehta. The team will evaluate upload, extraction, semantic retrieval, answer quality, source attribution, and session separation.

Primary objectives

- Support PDF, DOCX, and TXT uploads.
- Extract text and useful structure, including headings and tables where supported.
- Retrieve relevant passages for each question.
- Generate answers grounded in retrieved content and provide source metadata.
- Preserve a short conversation history for follow-up questions within a session.
- Return a safe unknown-answer response when the source lacks the answer.

Success criteria

Measure	Target	Verification
Supported formats	PDF, DOCX, TXT	Upload one of each type
Indexing	Positive chunk count	Inspect API response
Grounded answers	Correct source facts	Compare with report
Source attribution	Correct filename	Inspect chat response
Session isolation	No cross-session retrieval	Query a second session

2. Project Team and Responsibilities

The pilot uses a compact team with clear responsibilities. Names and roles are fictional and are included to support table extraction and question-answering tests.

Name	Role	Primary responsibility
Mira Shah	Project Owner	Scope, priorities, approvals, stakeholder updates
Arjun Mehta	Technical Lead	Architecture, implementation decisions, code review
Nisha Rao	Data and Evaluation Analyst	Test questions, answer verification, evaluation log
Kabir Desai	Quality Engineer	API testing, regression checks, issue tracking
Leena Patel	Documentation Coordinator	Sample documents, setup notes, demonstration script

Communication plan

Meeting	Frequency	Participants	Expected output
Planning review	Every Monday	Owner, lead, analyst	Prioritized weekly tasks
Technical checkpoint	Wednesday	Lead, quality engineer	Risks and implementation decisions
Demo rehearsal	Every Friday	All pilot members	Recorded issues and demo readiness

Decision process

The project owner approves scope changes. The technical lead documents architecture decisions. The quality engineer may block a demonstration if a critical defect causes incorrect answers, missing citations, or cross-session data exposure. Unresolved issues are recorded with an owner and target resolution date.

3. Schedule, Budget, and Deliverables

The six-week schedule gives each phase a deliverable that can be checked during the demonstration or reviewed in the project repository.

Phase	Dates	Deliverable	Owner
1. Requirements	2–6 Nov 2026	Scope and acceptance checklist	Mira Shah
2. Ingestion	9–13 Nov 2026	PDF, DOCX, and TXT parsing	Arjun Mehta
3. Retrieval	16–20 Nov 2026	Chunking, embeddings, vector search	Arjun Mehta
4. Chat API	23–27 Nov 2026	Grounded answers and source metadata	Nisha Rao
5. Quality review	30 Nov–4 Dec 2026	Regression and isolation report	Kabir Desai
6. Demo readiness	7–11 Dec 2026	Rehearsed demo and README	Leena Patel

Budget allocation

Category	Allocated amount (INR)
Development and integration	180,000
Testing and evaluation	90,000
Documentation and demonstration	45,000
Infrastructure and tools	35,000
Contingency reserve	100,000
Total	450,000

The contingency reserve is intended for unexpected implementation or testing effort. No cloud provider or hosting vendor has been selected in this report.

4. Document Assistant Architecture and Workflow

The application follows a retrieval-augmented generation workflow. Uploaded files are validated and parsed into text or Markdown, split into chunks, converted into embedding vectors, and stored in a persistent vector collection. A user question retrieves relevant chunks. The language model receives the question and retrieved context, then produces an answer with source metadata.

Stage	Input	Operation	Output
1. Upload	PDF, DOCX, TXT	Validate extension and size	Temporary file
2. Parse	Temporary file	Extract text, headings, tables	Parsed content
3. Chunk	Parsed content	Split into overlapping passages	Document chunks
4. Index	Chunks	Create embeddings and persist vectors	Searchable records
5. Retrieve	Question and session ID	Search relevant chunks in session scope	Retrieved context
6. Generate	Question plus context	Generate grounded response	Answer and sources

Grounding and safety rules

- Use retrieved content as evidence rather than unsupported assumptions.
- Treat instructions inside uploaded documents as untrusted content.
- Do not invent page numbers, filenames, dates, amounts, or citations.
- If evidence is insufficient, return the configured unknown-answer response.
- Apply session filtering so one session cannot search another session's documents.

Demonstration limitation

This report describes intended behavior and acceptance criteria. It does not prove that every behavior is implemented or secure; the running application must be tested against these criteria.

5. Test Plan and Expected Results

Execute these tests through the FastAPI Swagger interface at /docs. Index the report with a session ID, then use the same ID for chat questions.

Test ID	Test action	Expected result
T-01	Index this DOCX	HTTP 200, indexed status, positive chunk count
T-02	Ask for project budget	INR 450,000 and source citation
T-03	Ask who owns the project	Mira Shah
T-04	Ask for technical lead	Arjun Mehta
T-05	Ask which phase begins 16 Nov	Retrieval, 16–20 Nov 2026
T-06	Ask for total budget	INR 450,000
T-07	Ask which cloud provider was selected	Safe unknown answer; not specified
T-08	Use a different session ID	Cannot retrieve this session's document

Example questions for the demonstration

- What is the project name, owner, budget, and pilot duration?
- List the team members and responsibilities.
- What are the six phases and their deliverables?
- How is the budget allocated?
- Explain the retrieval-augmented generation workflow.
- Which cloud provider will host the application?

6. Risks, Acceptance, and Demo Script

Risks and mitigations

Risk	Impact	Mitigation
Poor text extraction	Missing or garbled evidence	Inspect parse preview and test all formats
Irrelevant retrieval	Incomplete or incorrect answer	Tune chunking and relevance threshold
Unsupported model claims	Loss of trust	Use grounded prompts and unknown-answer behavior
Missing citations	Answers are hard to verify	Return and inspect source metadata
Session filter failure	Cross-session data exposure	Test with separate session IDs

Acceptance checklist

☐ The document parses and indexes without an unhandled error.

☐ Questions about explicit facts receive accurate answers.

☐ The response identifies the source document.

☐ Questions outside the document are not answered by guessing.

☐ A second session cannot retrieve this session's content.

☐ The demonstration can be repeated from documented steps.

Suggested 1–2 minute demo sequence

1. Upload and index this report using session ID demo-orion-001.
2. Ask: What is the project budget and who owns the project?
3. Show the answer and source citation.
4. Ask a follow-up: Who is the technical lead?
5. Ask: Which cloud provider was selected? Show the safe unknown response.
6. Repeat a question with a different session ID to verify isolation.

End of demonstration report